

What is a shell program and what is a bash

A shell is a program that lets you run commands on your computer. Bash is a popular type of shell that has extra features like command history and auto-completion. Bash is commonly used in Linux because it's powerful and easy to use.

1. What is your home directory?

```
nachocortes02@Mac ~ % cd  
[pwd  
/Users/nachocortes02  
nachocortes02@Mac ~ %
```

2. What files/folders exist in it?

```
[nachocortes02@Mac ~ % ls -d ~/*/  
/Users/nachocortes02/Desktop/      /Users/nachocortes02/Output/  
/Users/nachocortes02/Documents/    /Users/nachocortes02/Pictures/  
/Users/nachocortes02/Downloads/    /Users/nachocortes02/Public/  
/Users/nachocortes02/HW3_files/     /Users/nachocortes02/R-Studio/  
/Users/nachocortes02/Library/       /Users/nachocortes02/cheat_sheet_files/  
/Users/nachocortes02/Movies/        /Users/nachocortes02/my_folder/  
/Users/nachocortes02/Music/  
nachocortes02@Mac ~ %
```

Where does the command `cd ../../` take you? Run the command `pwd` and explain the output!

The command `cd ../../` takes you two levels up in the directory structure, moving from the current directory to its grandparent directory.

The command `'pwd'` prints the working directory I am working currently. When I use the command `cd ../../` I move two levels up in the directory structure, so I see for example `/Users/nachocortes02/Downloads/`, running this command will take me to `/Users/nachocortes02/Desktop`

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1. What does the command ``cd`` do? Run the command ``pwd`` and explain the output!

```
[nachocortes02@Mac / % cd
[nachocortes02@Mac ~ % pwd
/Users/nachocortes02
nachocortes02@Mac ~ %
```

Running CD command without any arguments, it takes me to my home directory, and running pwd afterward will display the path of the current directory

Read the manual page of `ls`. What does the ``a`` flag do? What does the ``l`` flag do?

```
-a    Include directory entries whose names begin with a dot ('.').
      The number of blocks for the file
```

```
-l    (The lowercase letter "ell".) List files in the long format, as
      described in the The Long Format subsection below.
      Running CD command without any arguments, it takes me to my
```

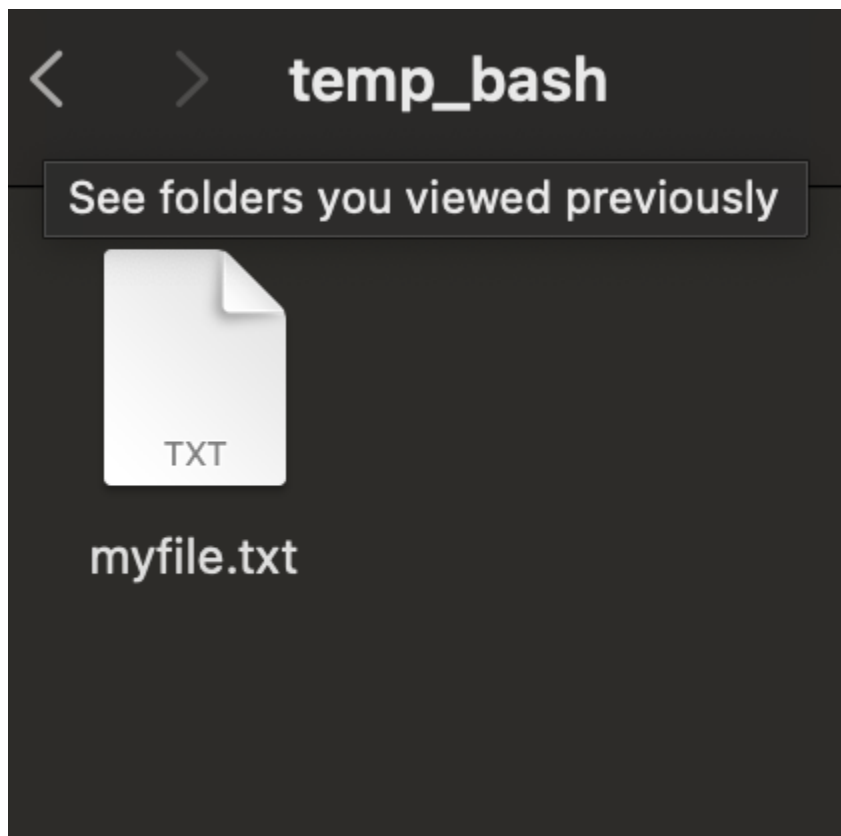
1. Create a folder within your home directory, which was identified in Question 2, and name it 'temp_bash'.

```
nachocortes02@Mac ~ % mkdir ~/temp_bash
nachocortes02@Mac ~ % ls ~
CV, Ignacio Cortés Niilus.pdf      Thesis .pdf
Desktop                            Trend Line of Business Expectations.pdf
Documents                          Untitled.pdf
Downloads                           cheat_sheet.html
HW3.html                           cheat_sheet.pdf
HW3.pdf                            cheat_sheet.qmd
HW3.qmd                             cheat_sheet_files
HW3_files                           e.pptx
Library                             e.qmd
Movies                              exam1_Ignacio_Cortés_Niilus.pdf
Music                               exam1_Ignacio_Cortés_Niilus.qmd
Output                              lab2.pdf
Pictures                            lab2.qmd
Public                              my_folder
R-Studio                           plot business exp.pdf
Resume 1. A.pdf                     temp_bash
Resume, Ignacio Cortés Niilus.docx
nachocortes02@Mac ~ %
```

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Create a new file using the command `touch` and name it `myfile.txt` inside the new folder `temp\_bash` and run `ls` to show that the file is inside the folder.

```
nachocortes02@Mac ~ % cd ~/temp_bash
nachocortes02@Mac temp_bash % touch myfile.txt
nachocortes02@Mac temp_bash % ls
myfile.txt
nachocortes02@Mac temp_bash %
```



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3. Run the `stat myfile.txt` command and explain the information retrieved from the output. Here is an example of what should be included in the output and a brief explanation for each part.

File Mode (-rw-r--r--)

This shows the file permissions: the owner has read and write access, while the group and others have read-only access.

Owner and Group (nachocortes02 staff)

Indicates the user (nachocortes02) and group (staff) that own the file.

Size (0)

This shows the file's current size in bytes e it's empty, so, it is 0 bytes).

Timestamps (Jan 26 13:28:57 2025)

These indicate the last access, modification, and change times of the file.

Blocks (0)

The number of disk blocks allocated to the file; since the file is empty, it uses no blocks.

IO Block (4096)

Represents the size of each block used for I/O operations (4KB in this case).

```
nachocortes02@Mac temp_bash % echo "This line is my first line." >> myfile.txt
nachocortes02@Mac temp_bash % echo "This line is my second line." >> myfile.txt
nachocortes02@Mac temp_bash % cat myfile.txt
]This line is my first line.
This line is my second line.
nachocortes02@Mac temp_bash % ]
```

```
nachocortes02@Mac temp_bash % cp myfile.txt copy_myfile.txt
nachocortes02@Mac temp_bash % ls
copy_myfile.txt myfile.txt
nachocortes02@Mac temp_bash %
```

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1. Explain the difference between `>` and `>>` based on the result of the Question 6.

```
nachocortes02@Mac temp_bash % echo "This line is a new line" > copy_myfile
nachocortes02@Mac temp_bash % cat copy_myfile.txt
This line is a new line
nachocortes02@Mac temp_bash %
```

This operator replaces the existing content in the file with the new content. If the file exists, its previous contents will be erased. It over writing instead of adding new content.